

Order Matching Engine — Technical Documentation

Overview

The **Order Matching Engine** is a Java Spring Boot + React application that simulates a simplified stock trading platform. It handles BUY and SELL orders using **FIFO** (price-time priority) and records trades asynchronously using **Kafka**. For more details on the Tech Stack and Local setup please check the README file.

Architecture

Frontend (React + Vite)

- Sends orders to backend
- Displays all orders and trade history

Backend (Java Spring Boot)

- Accepts and validates incoming orders
- Publishes them to Kafka
- Consumes and processes them with OrderMatchingService
- Matches orders and stores results in H2 (in-memory DB)

Kafka

- Topic: orders
- Acts as a decoupled message broker for asynchronous FIFO order processing

Order Matching Logic

- Incoming order is saved with status **PENDING**
- Order type is checked (**BUY/SELL**)
- The engine searches for matching counter orders:
 - For **BUY**: match against **lowest-priced SELLS** until the complete order cannot be matched further
 - For **SELL**: match against **highest-priced BUYs** until the complete order cannot be matched further
- Matches are made based on:
 - **Price**And if two orders have the exact same price then:
 - **Time priority**
- Status of an order is updated to **FILLED** or **PARTIALLY_FILLED**
- When two orders are matched then a trade is recorded with timestamp, symbol, price, quantity